IIT BOMBAY TECHFEST 2026–27 | NAMMASPACE CHALLENGE

🌌 NammaSpace Master Submission

Autonomous Monocular 3D Gaussian Splatting Digital Twin & Indoor Spatial Navigation Engine

Challenge: NammaSpace (Juspay)
Round: 1 (Reconstruction & Roaming)
Status: Production Ready (60 FPS)
Date: October 2026

Executive Overview: NammaSpace delivers an end-to-end framework transforming monocular smartphone video into an ultra-high-fidelity 3D Gaussian Splatting (3DGS) digital twin running universally in standard web browsers at locked 60+ FPS. Evaluators can explore three complete architectural environments encompassing **6,000,000+ splat primitives**, with Google Maps-style indoor landmark pins, real-time spatial search, and obstacle-aware A* pathfinding.

📂 ROUND 1 DELIVERABLES INDEX & DIRECT ACCESS LINKS

Deliverable Item	Direct Access Link / Location	Scope & Contents	Status
1. Live Web Application (Zero-Install WebGL Viewer)	nammaspace-digital-twin.vercel.app	Triple-Environment Showcase (Studio 1.13M, Atrium 1.73M, Courtyard 3.14M), 60 FPS, 3D pins, search, radar.	LIVE 60 FPS
2. Open-Source Code Repo (GitHub Repository)	github.com/Kaish-sheikh/...	Complete source code, Three.js shaders, A* pathfinder, mathematical ground leveling, training guide.	VERIFIED
3. Technical Abstract (Official PDF Document)	01_Technical_Abstract.pdf	Mathematical formulation, 3DGS radiance fields, tile rasterization, camera calibration, loop closure.	5 PAGES
4. Capture Protocol (SOP) (Standard Operating Procedure)	03_Standard_Operating_Procedure_SOP.pdf	3-loop capture checklist, optical conditioning, drift mitigation, and repeatability guide (20% rubric).	STANDARDIZED
5. Demonstration Video (3–5 Minute Walkthrough)	04_Demonstration_Video.mp4	Full video showing smartphone capture technique and real-time first-person interactive browser roaming.	INCLUDED

Deliverable Item	Direct Access Link / Location	Scope & Contents	Status
6. Evaluation Guide (Judges Reference Sheet)	`02_Web_Viewer_and_Project_Links.pdf`	One-click URL reference sheet, control scheme, environment specs, and troubleshooting notes.	**VERIFIED**

✨ KEY ARCHITECTURAL FEATURES & EVALUATION HIGHLIGHTS

🏛️ Triple-Environment 6.0M+ Showcase

Instant switching across Ultra-HD Studio (1.13M), Commercial Atrium (1.73M), and Spatial Courtyard (3.14M). All mathematically leveled to ground plane Y=0.

📍 Google Maps Indoor 3D Pins & Search

Dynamic 3D billboard pin markers with pulsing floor target halos and instant search bar filtering that illuminates matching world pins with keyboard Enter navigation.

🧭 Obstacle-Aware A* Waypoint Pathfinding

Projects an animated glowing dashed walking path on the floor avoiding obstacles, with real-time distance and estimated walking time calculations.

📡 Holographic 2D Tactical Floor Radar

Real-time synchronized canvas minimap showing player position, orientation delta chevron, dynamic 60° vision cone, and active waypoint blips.

Controls: `W` `A` `S` `D` Walk | `SHIFT` Sprint | `Mouse` Look | `ESC` / `Middle Click` Release Cursor | `V` Dollhouse Orbit | `T` Cinematic Tour